UMKC Impact & Translation Series

Why Commercialization Matters for Research Impact

Helping you move your work beyond the university

In today's research environment, success is no longer measured by publication alone. UMKC researchers possess the creativity and expertise to generate world-changing ideas. But impact does not stop at discovery.

Many researchers are ultimately motivated by impact - seeing their work improve lives, influence practice, or solve real problems beyond the university.

In many cases, that kind of impact requires translation. One pathway for doing that is commercialization: the process of moving research into real-world use through structured development, partnerships, or deployment.

This whitepaper explores the why, what, and how of commercialization at UMKC, emphasizing the opportunity and responsibility researchers have to bring innovation to the public through engagement with the Office of Commercialization.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **why commercialization is one pathway to impact.**

Introduction: Research That Reaches Beyond the Lab

Universities have long been engines of knowledge creation. But in today's interconnected and impact-driven research landscape, discovery alone is no longer sufficient. Public and private funders are now emphasizing societal benefit and seeking clear pathways from lab to life.

Commercialization, when done ethically and strategically, helps bridge this gap. It provides a structured route for bringing innovations to the public while supporting the broader mission of higher education: to advance knowledge *and* improve lives.

At UMKC, researchers in medicine, engineering, music therapy, the social sciences, and beyond are already embracing this pathway. They are not just publishing papers—they're creating startups, licensing technologies, and collaborating with industry to deliver meaningful outcomes.

How Translation Increases the Impact of Your Work

These benefits reflect what happens when research successfully moves into real-world use, whether through commercialization, partnerships, or other applied pathways.

1. Accelerated Societal Impact
Traditional dissemination through journal articles reaches other scholars, but it often falls short of creating real-world change. Commercialization brings discoveries out of the lab and into the hands of the people who need them.

Examples of impact:

- **Medical devices** that deliver better care to underserved patients
- **AI algorithms** applied to climate risk or public health
- **Tools and therapies** made accessible through commercial distribution

As emphasized in *EMBO Reports*, the societal impact of research can be difficult to measure, but it is increasingly demanded by policymakers and funders. Even basic science, once seen as untouchable in its purity, is being reevaluated through the lens of public benefit.

2. Scientific Responsibility and Societal Ethics
Commercialization isn't just a strategic move - it's a moral imperative. As Stephanie J. Bird argues in *Socially Responsible Science Is More than "Good Science"*, scientists hold a collective responsibility to consider the foreseeable impacts of their research. Society expects more than technical rigor; it expects relevance, equity, and foresight.

Commercialization helps fulfill this broader ethical commitment by:

- Bringing public investments in science full circle through societal returns
- Ensuring that innovations do not stay siloed in academia
- Encouraging scientists to engage in ethical foresight and policy dialogue

3. Enhanced Research Visibility and Value
Engaging in commercialization draws new attention to your work. Researchers who develop translational pathways often see:

- Invitations to participate in applied research centers
- Features in Whiteboard2Boardroom videos and innovation showcases
- Competitive advantages in funding proposals, including SBIR/STTR grants and translational research calls

UMKC's Office of Commercialization ensures that your innovations are visible, evaluated, and strategically promoted.

4. Economic Growth and Entrepreneurial Opportunity
By commercializing, faculty contribute directly to Kansas City's economic ecosystem. Pathways include:

- Launching startups that create local jobs
- Earning licensing revenue that supports labs and graduate students
- Accessing funding from entrepreneurial programs like Digital Sandbox KC

Your innovation doesn't just serve society - it fuels it.

5. Stronger Industry Partnerships
Commercial-ready research attracts interest from companies, non-profits, and government agencies, leading to:

- Joint research agreements
- Industry-led beta testing
- Early adoption in applied settings

UMKC's growing partnerships in fields like health tech, engineering, and education are often seeded by faculty innovation.

Understanding the Hesitations: What Holds Researchers Back?

Despite these benefits, national studies and faculty interviews consistently show a few recurring challenges:

"I'm not a businessperson." - You don't have to be. The Office of Commercialization provides business modeling, IP support, and startup guidance. Many researchers stay focused on the science while partnering with entrepreneurs and staff experts.

"I don't have time." - You already invest years developing your discoveries. A 30-minute conversation with our team can clarify whether a path to market exists. The earlier you engage, the smoother the process.

"Isn't commercialization a distraction from academic rigor?" - No. In fact, commercialization can strengthen your academic output. Studies show that faculty who patent or license technologies often have higher publication productivity than their peers.

Guiding Question:
Where could your work make a meaningful difference beyond the university, and what would need to happen for that impact to be realized?

How the Office of Commercialization Supports Your Impact

Within the broader landscape of research impact and translation, the Office of Commercialization serves as a key partner in helping faculty move their work toward real-world use.

The UMKC Office of Commercialization is your launch point for bringing innovation to life. We work alongside faculty and research teams to move discoveries from concept to community through licensing, startup creation, or strategic partnerships. Our mission is to help you maximize the societal relevance, visibility, and long-term value of your research.

From intellectual property protection to market navigation, our office provides expert guidance and hands-on support. Whether you're submitting your first invention disclosure or refining your go-to-market strategy, we're here to help you take the next step with confidence.

- **Invention Discovery & Protection**
  - Submit an **Invention Disclosure Form (IDF)** as soon as your idea shows signs of novelty.
  - Receive expert evaluation of market fit, value proposition, and commercialization strategy.
  - Get tailored guidance on protecting your innovation before public disclosure or publication.
- **Market Research & Pathfinding**
  - Conduct market assessments to identify needs, competitors, and target users.
  - Map potential industry partners, licensees, and application pathways.
  - Evaluate startup feasibility, including business models and alignment with funding opportunities.
- **Licensing & Startup Support**
  - Navigate licensing agreements with legal and business development support.
  - Access mentorship and connections to technical, regulatory, and business advisors.
  - Define founder roles and build effective teams for spinout companies.
  - Identify early-stage funding sources including SBIR/STTR and local accelerator programs.
- **Institutional Advocacy**
  - Showcase your innovation through pitch events, media coverage, and innovation showcases.
  - Build connections with funders, industry partners, and regional collaborators.

To explore these services and more, visit umkcinnovates.com/tech-commercialization, or contact us directly to start a conversation.

What Commercialization Looks Like in Practice at UMKC

The following examples illustrate how research at UMKC moves toward impact through translation and commercialization in practice.

Case Study: Advancing Disease Detection Through Licensing

Innovation: Lectin-based detection of glycoRNA
Researcher: Dr. Mingui Fu, UMKC School of Medicine
Field: Biomedical Diagnostics

Dr. Mingui Fu developed a novel approach to detecting glycoRNA using lectin-based methods—offering the potential for dramatically simpler and more accessible testing for cancer, immune disorders, and infectious diseases. Recognizing the translational potential of this breakthrough, Dr.

Fu submitted an invention disclosure to the Office of Technology Transfer.

From there, the Office of Commercialization partnered closely with Dr. Fu to chart a path toward real-world application. This included:

- Conducting a market assessment to identify high-need sectors and relevant commercial applications
- Identifying and prioritizing the most promising licensing targets
- Creating professional marketing materials to promote the technology
- Proactively contacting and engaging potential licensees

The Office is currently leading licensing discussions and working to finalize material transfer agreements with two well-qualified industry partners. These agreements will allow the companies to evaluate the technology and serve as a key step toward securing a full licensing agreement.

Impact Outlook:
Dr. Fu's innovation has the potential to lower the cost and complexity of biomarker detection across multiple medical domains. By engaging early and proactively with the commercialization team, Dr. Fu maximized his invention's visibility and accelerated its pathway to evaluation by industry.

Case Study: From Lullabies to Lung Health

Innovation: In-bassinet music-reward device for preterm infants
Researcher: Dr. Amy Robertson, UMKC Conservatory (in collaboration with Children's Mercy Hospital)
Field: Music Therapy, Neonatal Medicine

Dr. Amy Robertson's research on music therapy revealed that premature infants perceive tone and melody as a form of positive reward. Motivated by the potential clinical implications, she asked a powerful question: could music be used as a reinforcement tool to encourage spontaneous breathing in NICU infants?

Working alongside physicians at Children's Mercy Hospital's NICU, Dr. Robertson developed a hypothesis: playing a short music clip immediately after a spontaneous breath might prompt more voluntary breathing, reducing infants' time on mechanical ventilation and the associated medical risks.

However, the concept presented a logistical challenge—deploying live musicians in the NICU was not feasible. To scale the idea, she envisioned an automated in-bassinet device capable of detecting spontaneous breaths and delivering an audio reward.

Dr. Robertson brought this concept to the UMKC Office of Commercialization, where we helped her refine and define the scope of her innovation. Our team provided:

- Design and strategy support to translate her concept into an actionable development plan
- Connections to industry advisors to consult on device feasibility and sensor integration

- Proposal support for the Entrepreneurship Innovation Grant

With this guidance, Dr. Robertson successfully secured a $10,000 EIG award to fund early development. We are now helping her explore collaborations with ventilator device manufacturers to create a working prototype and validate the concept in clinical settings.

Impact Outlook:
If successful, this innovation could represent a breakthrough in neonatal care—combining behavioral science, music therapy, and medical device engineering to reduce dependency on ventilation in premature infants. Dr. Robertson's creativity, paired with UMKC's commercialization support, is transforming a research insight into a potentially life-saving tool.

Getting Started: Action Steps for UMKC Researchers

If you're working through the UMKC Impact & Translation Series, these are the natural next steps to begin applying what you've learned:

1. **Talk with Us Early -** Before you publish or present findings that could be commercially relevant, contact the Office of Commercialization. We'll help you understand your options.
2. **Submit an Invention Disclosure -** This form kicks off internal evaluation and starts the protection process.
3. **Explore Opportunities -** Join community speaker series or commercialization workshops to learn more about available resources.
4. **Consider All Pathways -** Not every innovation becomes a startup. Licensing, nonprofit deployment, open science models, and hybrid approaches are all supported.
5. **Build a Team -** Work with the Office and partners like Whiteboard2Boardroom, Digital Sandbox, or entrepreneurial students to strengthen your project's success potential.

Conclusion: Your Ideas Deserve to Go Further

This is not about changing the nature of your work. It is about expanding its reach.

Your research already has value. The question is how that value might move beyond the university and what it would take to get it there.

The UMKC Impact & Translation Series is designed to help you think more clearly about that path. The Office of Commercialization is here to help you explore it.

Resources & Contacts

UMKC Office of Commercialization

📍 Website: www.umkcinnovates.com/tech-commercialization
📧 Email: techcomm@umkc.edu
📞 Phone: 816-235-6373

Additional Resources

- Invention Disclosure Form (IDF)
- Digital Sandbox KC
- Whiteboard2Boardroom Program
- SBIR/STTR Funding Advising